

Part: **3.4 – STRENGTH OF COMPOSITE**

Chapter: **3468**

Title: **POST-BUCKLING OF COMPOSITE PANELS**

Revision: **0.1**

Date: **02/09/2021**

1 INTRODUCTION

This chapter proposes an approach to deal with post-buckling in rectangular plates, or (slightly) curved shells, made of composite materials, in particular during initial phases of the design / sizing process.

Post-buckling is a very complex phenomenon and the methodology presented herein is quite straightforward/simple, hopefully approximated, but not intending to predict accurate margins but just a method to mitigate the risk of bad initial designs before a final reliable validation.

In case post-buckling is permitted in a composite panel of some structural significance, dedicated testing should be mandatory before completing the qualification/certification of the part (plus, typically, detailed numerical simulation analysis).

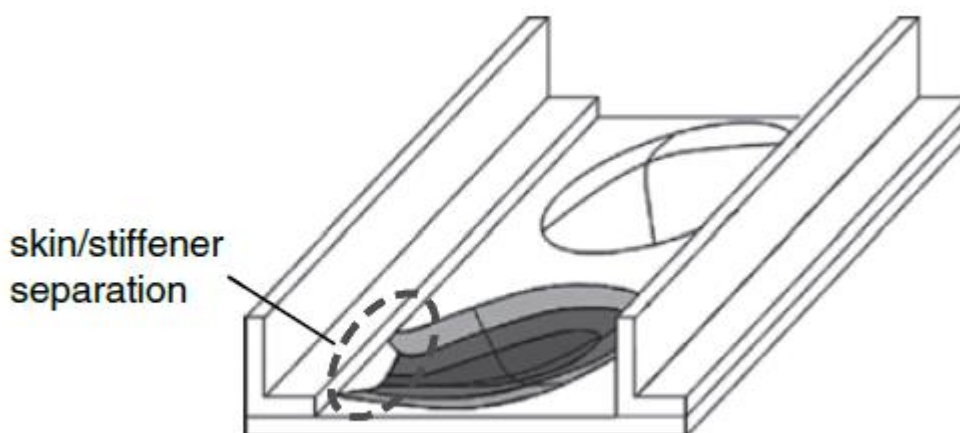


Figure 1. Post-buckled skin between stiffeners in a stiffened panel (Ref. 2)

2 METHOD DESCRIPTION

Calculation of the static margins of the post-buckled structure shall be done by two strategies:

- Using some familiar straightforward methodologies, adapted from isotropic materials, i.e. the effective-skin concept or the diagonal tension theory
- Limiting the excess of load over the buckling onset, based on previous testing experience, by means of the post-buckling-cut-off ratio

2.1 POST-BUCKLING STRENGTH

2.1.1 Post-buckling under compression

The post-buckled ultimate strength of a rectangular narrow plate/shell under compression will be estimated using the traditional concept of the effective skin width.

Details for the calculation of this magnitude can be found in chapter 3485 "EFFECTIVE SKIN WIDTH".

The updated values of the load and strain values for the reduced panel width shall be used to obtain the static margins later on.

2.1.2 Post-buckling under shear

A simplified version of the diagonal tension theory for panels made of isotropic materials can be used for this purpose.

Details about the recalculation of the panel loads, and the induced forces in the stiffening members composing the panel frame, can be found in chapter 3460 "SHEAR WEBS". A couple of remarks:

- For calculating the diagonal tension factor, consider the buckling level as if exclusively subjected to the shear load component
- Assume for simplicity a diagonal tension angle of 45°

2.1.3 Static margins of the post-buckled panel

Proceed as follows:

- Get the strength reserve factor of the panel laminate (plain strength or damage tolerance as applies) with the revised panel flows calculated with the diagonal tension analysis.
- Update the axial loads on the stiffening members composing the panel frame (as skin stringers) resulting after the effective-skin and diagonal tension analyses. Then, get the static margins associated to this element (plain or damage tolerance strength, crippling and global/columns buckling)

2.2 POST-BUCKLING CUT-OFF

The post-buckling cut-off ratio is defined as the division of the expected ultimate strength of the (post-buckled) panel and its buckling onset level, i.e.:

$$PBCO = F_{failure} / F_{buckling}$$

This ratio must be conveyed in advance as a limit of the ultimate strength, in order to cover imprecisions of the methodology discussed previously, and/or failure modes not considered (eg. stringer disbond). Its value shall be preferable established from tests sufficiently representative of the structure object of analysis. Otherwise, the following law is generally recommended for narrow rectangular plates and shells with low curvature (*)(**).

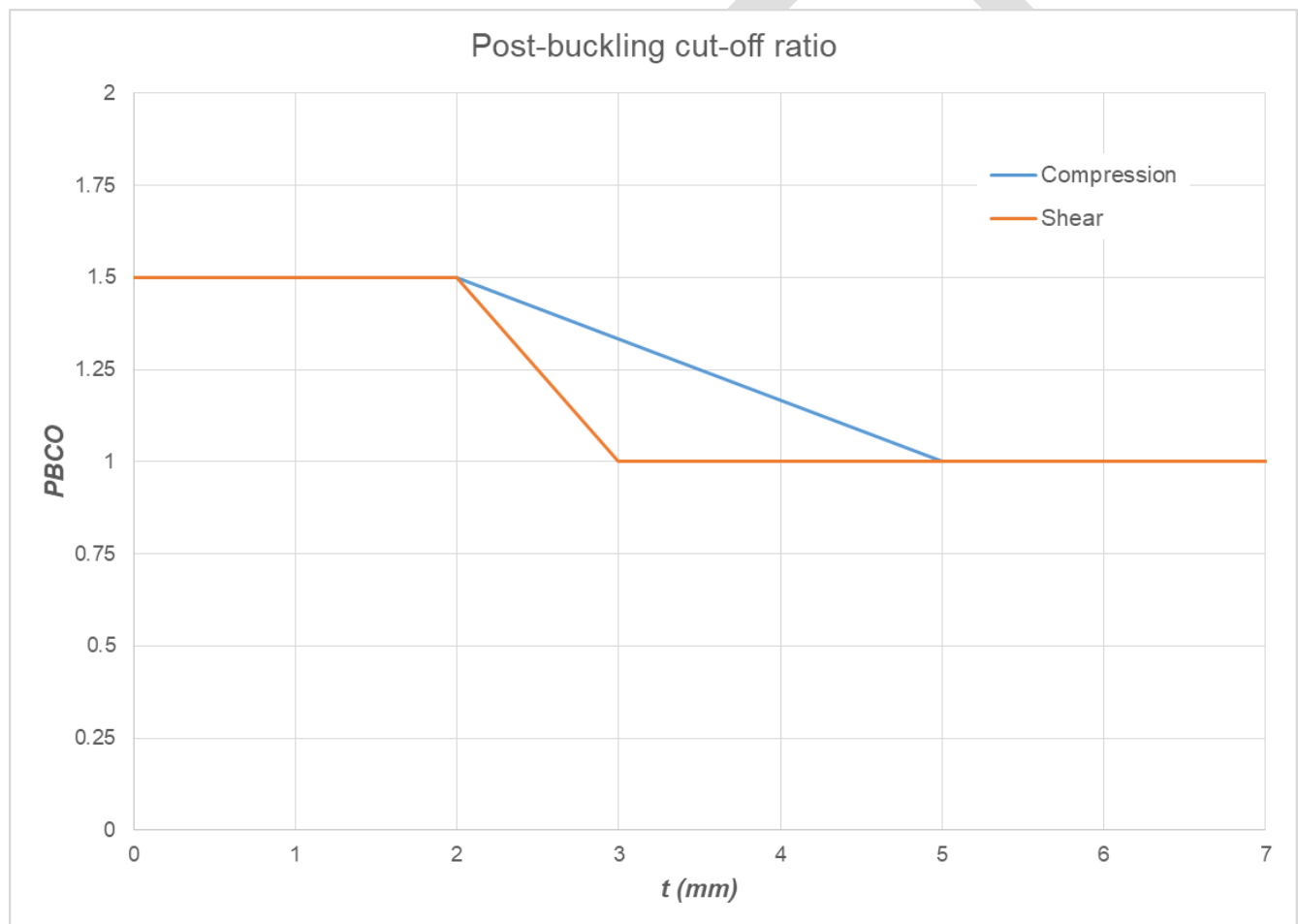


Figure 2. Default values of the post-buckling cut-off ratios

(*) Narrow plate/shell is considered if the length is greater than the width. For wider plates/shells, post-buckling in compression should not be permitted.

(**) Design values for the PBCO ratio shall be confirmed or reviewed as eventual new panels tests data are collected and processed. Values in graph have been inferred from a very limited set of legacy tests, which are

summarized in Ref. 1 (this proposal must be conservative for panels up to 2 mm thickness, whereas the few tests data available at 3mm thickness have been used to complete the graph by interpolation/extrapolation).

If the failure indices associated to post-buckling in compression and in shear, when considered separately, are:

$$R_C = 1 / (PBCO_C \cdot BL_C)$$

$$R_S = 1 / (PBCO_S \cdot BL_S)$$

...where the buckling levels are the ratios between the buckling load and the (ultimate) applied load:

$$BL = F_{buckling} / F_{applied}$$

Then the proposed formula to calculate the reserve factor under combined loads is the same expression used for calculating the buckling level under combined loads:

$$RF = 2 / [R_C + (R_C^2 + 4 \cdot R_S^2)^{0.5}] \quad (*)$$

(*) See chapter 3450 "PLATE/SHELL BUCKLING"

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A/C		Aircraft (style "Abbreviation")
BL	-	Buckling Level, ratio between buckling load and applied (ultimate) load
C	-	As subscript, relative to compression load
S	-	As subscript, relative to shear load
RF	-	Reserve Factor
TBC		To Be Confirmed

REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 TEA-T-NT-210048	Is. A	Manual Teórico del Programa ARPA
Ref. 2 [Christos Kassapoglou]	2 nd Ed.	DESIGN AND ANALYSIS OF COMPOSITE STRUCTURES

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 02/09/2021	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEPAP-TL4)	G. Baños 02/09/2021	F. Glock DD/MM/2021

Table 3. Approval signatures table for last revision of this chapter

DRAFT